Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. We trained each model for 50 epochs with a batch size of 128.

2 Results

Figure 1 shows the accuracy by dataset.

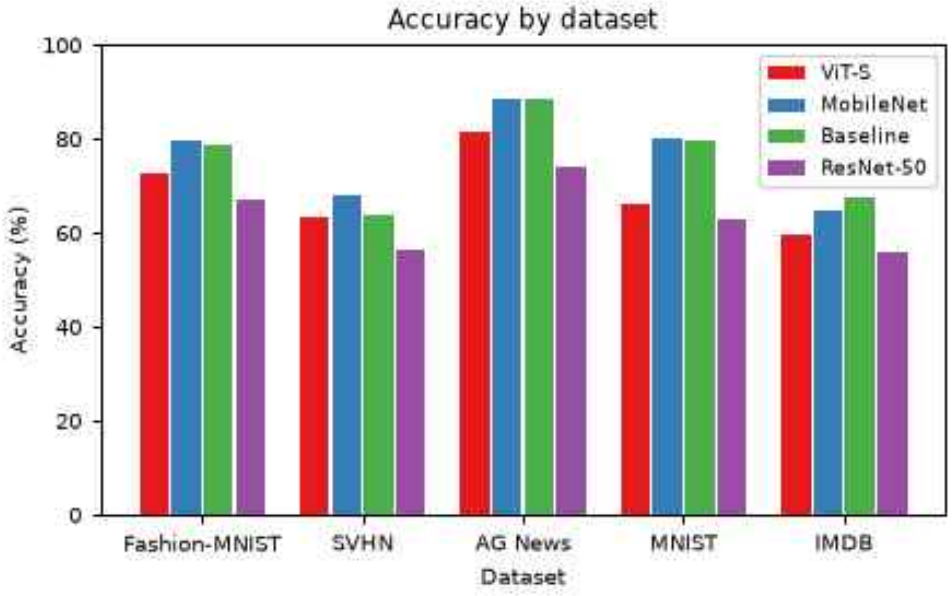


Figure 1: Accuracy by dataset for the models in this experiment.

ResNet-50 reaches 73.1% accuracy on MNIST (Figure 1). MobileNet outperforms ResNet-50 by 9 percentage points on IMDB (Figure 1). On Fashion-MNIST, Baseline obtains a accuracy of 128.6%. Baseline reaches 81.5% accuracy on AG News.

Figure 2 shows the accuracy during training.

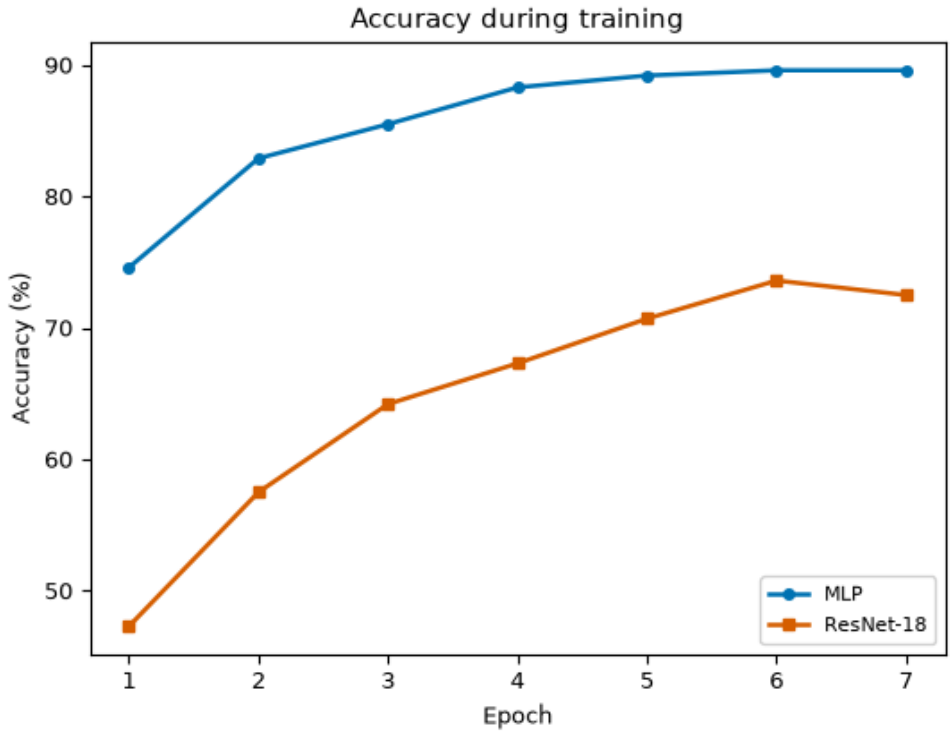


Figure 2: Accuracy during training for the models in this experiment.

ResNet-18 attains a accuracy of 38.0% at epoch 1. As shown in Figure 2, MLP outperforms ResNet-18 by 25.4 percentage points at epoch 2.

3 Discussion

On STL-10, XGBoost obtains an accuracy of 79.8%. The differences between the models are discussed in the next section.